1st August 2025

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To: *PLoS Biology* editorial board.

Dear editors,

Please find attached our manuscript entitled "Modeling the influences of non-local connectomic projections on geometrically constrained cortical dynamics" that we are submitting for publication as a *Research Article* in *PLoS Biology*.

A central question in neuroscience is how the structural wiring of the cortex gives rise to its intricate functional dynamics. The prevailing view in the field—that cortical dynamics stem from the complex web of fibers connecting distributed cortical networks—has unpinned decades of research and driven massive global investments in connectome mapping. But this view was recently challenged by Pang et al., *Nature* (2024), whose striking findings demonstrated that **many aspects of cortical dynamics, as measured using fMRI, can be accurately captured by *ignoring the connectome*.** Their model made the dramatic simplification of instead incorporating *purely geometric constraints* on brain dynamics—as the propagation of waves along the cortical surface. The results of this high-profile paper upturned the field of macroscopic neuroscience, leaving a major open question for the field to reconcile: *why are geometric models of brain dynamics so successful when they neglect fast-conducting, long-range projections that facilitate rapid communication between distant brain regions*?

Our paper develops a novel mathematical model of macroscale cortical dynamics to provide the first explanation of this major open question.

Our new model allows us to disentangle the contributions of both: (i) geometrically constrained dynamics (mediated by a continuous cortical sheet); and (ii) non-local projections that are wired independently of the sheet's geometry. Our comprehensive range of results—spanning systematic investigations of targeted stimulation-response and spontaneous dynamics—support a more nuanced picture of cortical dynamics that explains why the geometric approximation can be so accurate under specific circumstances. Specifically: while non-local connectomic projections are essential for shaping rapid, stimulus-evoked responses (on millisecond timescales), their contribution diminishes for spatially spread inputs (as in spontaneous dynamics) and on slower timescales (as measured by fMRI), in which a simpler geometric model becomes a surprisingly powerful approximation.

Supported by a comprehensive set of novel modelling results, our work establishes a fundamentally new picture of the function of cortico-cortical interactions, in which fast long-range fibre bundles are crucial for the rapid processing of (e.g., sensory) information across distributed cortical networks; while geometric constraints are sufficient to capture slower, spontaneous dynamics. As our work reconciles a major and topical open question in macroscale human neuroscience, we believe it will have a major impact and be of great interest to the readership of *PloS Biology*.

Thank you for considering our submission. Sincerely,



Dr Ben D. Fulcher, on behalf of the authors.